TAI NGUYEN PHU

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Final-year Computer Science student with a strong background in Machine Learning and Natural Language Processing (NLP), currently specializing in Large Language Models (LLMs) and generative AI. Seeking an AI/NLP Engineer internship or entry-level position focused on the development, optimization, and deployment of LLM- and generative AI-based applications.

EDUCATION

Bachelor of Computer Science — **Sep 2022 - Jun 2025**
University of Information Technology - VNUHCM, Vietnam — **GPA:** 8.24/10

CERTIFICATIONS

IELTS Academic (Overall: 7.0) — **Feb 2022**

PROJECTS

Vietnamese Legal Document Retrieval — **Apr 2025 - Apr 2025**

Technologies: Python, PyTorch, SentenceTransformers, FAISS, Gradio, Docker, Pandas

- Fine-tuned a multilingual Sentence-BERT model on a curated Vietnamese legal corpus, achieving domain-specific semantic representation for accurate document retrieval.
- Boosted retrieval performance dramatically, achieving NDCG@10: 60.4% and MAP@10: 53.6% on the MTEB benchmark (BKAI Legal Retrieval dataset), compared to only 0.023%/0.014% with the pretrained model.
- Designed a full semantic retrieval pipeline including data preprocessing, model fine-tuning, evaluation with MTEB benchmark, and real-world deployment.
- Built a GPU-accelerated FAISS index enabling sub-second approximate nearest neighbor (ANN) search across 100k+ legal documents.
- Developed a user-friendly Gradio web interface for semantic search queries, fully containerized with Docker for seamless deployment.

Semantic Book Recommender — **Mar 2025 - Apr 2025**

Technologies: Python, PyTorch, HuggingFace Transformers, SentenceTransformers, Gradio, Pandas, NumPy

- Developed a semantic-based book recommendation system leveraging sentence embeddings and large language models (LLMs) for context-aware, personalized suggestions.
- Implemented vector search using pre-trained sentence embeddings to find semantically similar books beyond keyword matching.
- Applied zero-shot classification via pre-trained LLMs to categorize books into genres without requiring labeled datasets.
- Conducted sentiment analysis on user reviews to enhance recommendation quality by factoring reader opinions.
- Built an interactive Gradio dashboard enabling users to input preferences and receive tailored recommendations in real time.

Job Application AI Assistant — **Mar 2025 - Mar 2025**

Technologies: Python, PyTorch, Llama, LangChain, Groq API, FAISS, Streamlit

- Developed an AI-powered assistant utilizing Llama 3.3-70B via Groq API and LangChain to generate personalized cold emails and job application strategies.
- Integrated Groq API for ultra-fast LLM inference, enabling low-latency, high-throughput real-time interactions.
- Implemented a semantic search system using FAISS to match user profiles with relevant job postings and enhance recommendation accuracy.
- Built and deployed an interactive Streamlit web application supporting real-time, context-aware job application assistance.

Scene Text Recognition — **Feb 2025 - Mar 2025**

Technologies: Python, PyTorch, YOLOv11m, OpenCV, CRNN, FastAPI, Ray Serve, Streamlit

- Built an end-to-end OCR pipeline combining YOLOv11m for scene text detection and CRNN (ResNet34 backbone) for sequence-based recognition.
- Trained and fine-tuned models on the ICDAR2003 dataset, achieving ~88% precision in real-world scene text detection tasks.
- Deployed scalable, high-performance API services using FastAPI and Ray Serve, optimized for GPU acceleration.
- Developed a modular Streamlit web application for real-time text detection and easy integration into external systems.

Data Science Job Salary Prediction — **Dec 2024 - Jan 2025**

Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

- Developed a machine learning-powered salary prediction model for Data Science jobs, trained on real-world Glassdoor job market data.
- Engineered and optimized features through exploratory data analysis (EDA), encoding, and feature importance evaluation.
- Trained and compared multiple models (XGBoost, Random Forest, Decision Tree, Gradient Boosting, Linear Regression, SVM), achieving $R^2 = 0.815$ with XGBoost. Maximize model performance with hyperparameter optimization using GridSearchCV.
- Built and deployed an interactive Streamlit web application, enabling users to input job attributes and receive real-time salary predictions.

SKILLS

Languages: Python, C++, SQL

Frameworks: PyTorch, Transformers, LangChain, FAISS, OpenCV, SciKit-Learn, XGBoost, Numpy, Pandas

Deployment: Git, Docker, AWS (basic), FastAPI, Ray Serve, Weights & Bias, Streamlit, Gradio

Soft Skills: English, Presentation, Critical Thinking, Teamwork, Problem Solving